

VELOCITY

ARS + Transaction Analytics

How We Measure

Methodology & Data Integrity

Every client-facing rate, the population it is measured over, and the story of how we made the headline numbers defensible.

JUNE 2026

STRICTLY CONFIDENTIAL

THE STORY

A headline rate that didn't reconcile

The symptom.

Our take rate led the deck at ~30%. The validated notebook put the same portfolio near 80%. A 50-point gap on the first number a client sees is a credibility problem.

~30% → ~80%

portfolio DCTR, once measured
over the right population

The cause.

Take rate divided by the eligible book — open accounts narrowed by status and product code — instead of by all open accounts. The denominator, not the math, was the bug.

The fix.

We codified one canonical base per metric. The portfolio rate divides by all open accounts; the eligible cut is kept as a labeled companion, shown side by side.

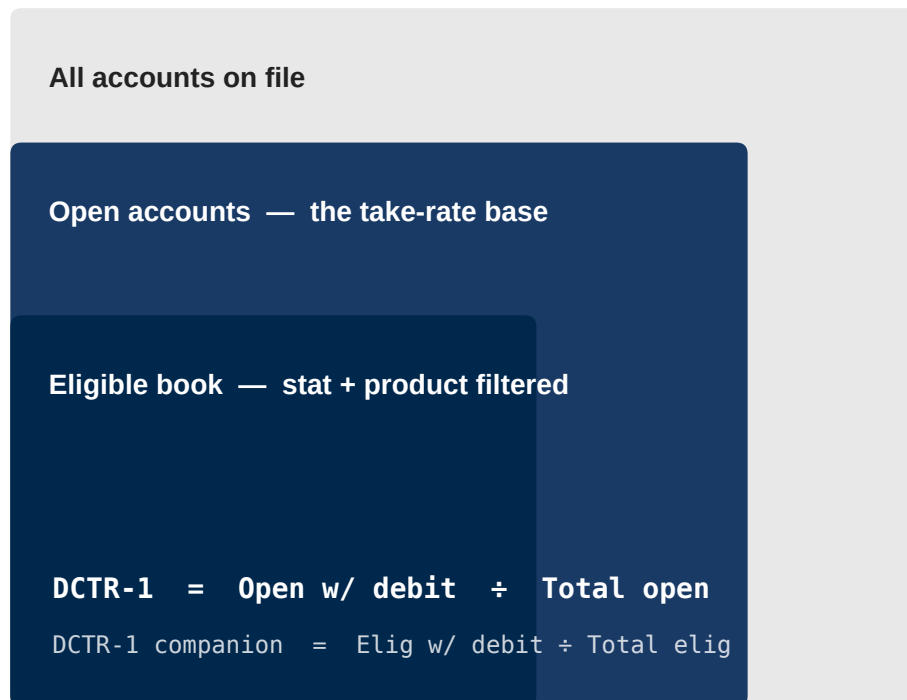
The safeguard.

Every published rate writes to a per-run audit file with its denominator and count. Two runs diff in one command; any rate that moves more than a point flags itself.

THE FRAMEWORK

One canonical base per metric

Rates differ only by the population beneath them. We name that population once, so every number reconciles to a known base.



WHICH BASE EACH SLIDE USES

Portfolio headline

All open accounts

DCTR-1, DCTR-3

Methodology comp

Open + Eligible, shown side by side

DCTR-2, companions

Eligible detail

Eligible book only

DCTR-4/5, 9, A7.x, overlays

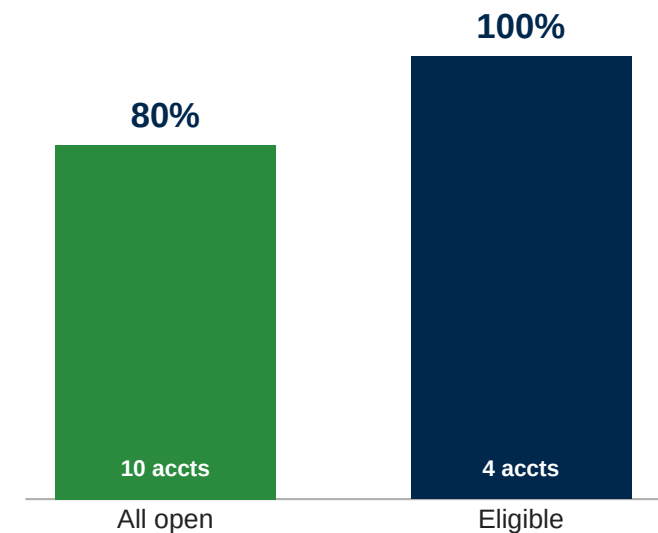
Detail breakdowns stay on the eligible book on purpose: a branch- or balance-tier take rate answers “adoption within the program,” not “across the whole portfolio.”

Debit Card Take Rate

The numerator is identical everywhere — debit_mask counts the cards. Only the population beneath it changes.

SLIDE	FORMULA	
DCTR-1	open w/ debit ÷ total open	Open
DCTR-1	elig w/ debit ÷ total elig	Eligible
DCTR-2	the two rates, side by side	Both
DCTR-3	open w/ debit ÷ total open (TTM)	Open
DCTR-3	elig w/ debit ÷ total elig (TTM)	Eligible
DCTR-4/5	debit ÷ total	Eligible
DCTR-9 · A7.x	debit ÷ total	Eligible

Worked example: same 8 cards, two bases



Narrowing to eligible products raises the rate — which is why the portfolio headline uses the open base.

REFERENCE

Calculations by section

Every shipped rate, its formula, and the population it divides by.

Overview

eligibility_rate

eligible ÷ all accounts

drop-off %

(prev – curr) ÷ prev stage

personal / business

segment ÷ eligible

Reg E

opt-in rate

opted-in ÷ eligible personal

(base)

eligible personal, not debit-narrowed

NaN handling

missing → not opted in

Attrition

overall_rate

closed ÷ all rows

l12m_rate

l12m closed ÷ open at start

revenue lost

last_spend × ic_rate × 12

Value

delta

rev/acct with – without debit

pot_hist

awo × delta × DCTR-1 rate

note

tracks the portfolio base

Mailer

resp_rate

responders ÷ mailed

penetration

responders ÷ eligible w/ card

Insights

S1 gap

accts_without × delta

realistic

total_gap × 0.25

S4–S8

inherit upstream bases

THE SAFEGUARD

Numbers that defend themselves

Each run writes `rates_audit.csv` — one row per published rate, with the exact denominator label and count.

slide_id	metric	value	denominator_n	denominator_label
DCTR-1	overall_dctr	0.8100	12,000	open accounts
DCTR-1	eligible_dctr	0.3200	4,800	eligible accounts
DCTR-2	open_dctr	0.8000	12,000	open accounts
DCTR-3	l12m_dctr	0.7400	1,950	open accounts (L12M)
A9.1	overall_rate	0.0600	12,000	all rows
A11.1	pot_l12m	412,500	—	active personal w/o debit

Diff two runs and every line is a rate that moved. Intended change, or regression — you know before the deck reaches the client.

Accurate first. Then beautiful.

`shared/brand.py` · `rates_audit.csv`